

Type Evaluation / Verification Report

Instrument Under Test		Test Conditions	
Manufacturer	Sample Scales Ltd.	Test Location	Demo Lab, Bay 2
Model	SS-150	Tested On	2026-09-26T08:42:55+00:00
Serial No.	SN-03-001	Tested By	Rahul Verma
Accuracy Class	III	Temperature	23.5 °C
Max Capacity	15000.0 g	Humidity	48.0 %RH
Min Capacity	100.0 g	Pressure	1008.0 hPa
Scale Interval (e)	5.0 g		
Verification Scale Interval (d)	5.0 g		
Max Tare	0.0 g		

⚠ **Rounding-trap detected.** The naive calculation and the R 76-1 changeover-point method disagree on the verdict for at least one reading below. The R 76-1 result is authoritative.

Test Results

Test	Load (g)	Error (g)	MPE (±g)	MPE Used	Naive rounding	Result	Clause	Explanation
weighing	100.0	0.0	2.5	0%	0.0 → PASS	PASS	R76-1 A.4.4; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±2.5 g (load = 20e, band 0e to 500e, MPE = 0.5e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 100 + 2.5$, corrected by $E0 = +0$.
weighing	2500.0	0.5	2.5	20%	0.0 → PASS	PASS	R76-1 A.4.4; R76-1 3.5.1, Table 6	Error +0.5 g vs MPE ±2.5 g (load = 500e, band 0e to 500e, MPE = 0.5e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 2500 + 2.5$, corrected by $E0 = +0$.
weighing	2505.0	-0.5	5.0	10%	0.0 → PASS	PASS	R76-1 A.4.4; R76-1 3.5.1, Table 6	Error -0.5 g vs MPE ±5 g (load = 501e, band 500e to 2000e, MPE = 1e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 2505 + 2.5$, corrected by $E0 = +0$.
weighing	5000.0	7.0	5.0	140%	5.0 → PASS	FAIL	R76-1 A.4.4; R76-1 3.5.1, Table 6	Error +7 g vs MPE ±5 g (load = 1000e, band 500e to 2000e, MPE = 1e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 5005 + 2.5$, corrected by $E0 = +0$. Rounding trap: the naive reading (indication - load = +5 g) would have given PASS.
weighing	6200.0	0.0	5.0	0%	0.0 → PASS	PASS	R76-1 A.4.4; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±5 g (load = 1240e, band 500e to 2000e, MPE = 1e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 6200 + 2.5$, corrected by $E0 = +0$.
weighing	7500.0	0.5	5.0	10%	0.0 → PASS	PASS	R76-1 A.4.4; R76-1 3.5.1, Table 6	Error +0.5 g vs MPE ±5 g (load = 1500e, band 500e to 2000e, MPE = 1e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 7500 + 2.5$, corrected by $E0 = +0$.
weighing	10000.0	-0.5	5.0	10%	0.0 → PASS	PASS	R76-1 A.4.4; R76-1 3.5.1, Table 6	Error -0.5 g vs MPE ±5 g (load = 2000e, band 500e to 2000e, MPE = 1e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 10000 + 2.5$, corrected by $E0 = +0$.
weighing	10005.0	1.0	7.5	13%	0.0 → PASS	PASS	R76-1 A.4.4; R76-1 3.5.1, Table 6	Error +1 g vs MPE ±7.5 g (load = 2001e, band 2000e to 10000e, MPE = 1.5e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 10005 + 2.5$, corrected by $E0 = +0$.
weighing	12500.0	0.0	7.5	0%	0.0 → PASS	PASS	R76-1 A.4.4; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±7.5 g (load = 2500e, band 2000e to 10000e, MPE = 1.5e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 12500 + 2.5$, corrected by $E0 = +0$.
weighing	15000.0	0.5	7.5	7%	0.0 → PASS	PASS	R76-1 A.4.4; R76-1 3.5.1, Table 6	Error +0.5 g vs MPE ±7.5 g (load = 3000e, band 2000e to 10000e, MPE = 1.5e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 15000 + 2.5$, corrected by $E0 = +0$.
weighing	15000.0	0.5	7.5	7%		PASS		

Test	Load (g)	Error (g)	MPE (±g)	MPE Used	Naive rounding	Result	Clause	Explanation
					0.0 → PASS		R76-1 A.4.4; R76-1 3.5.1, Table 6	Error +0.5 g vs MPE ±7.5 g (load = 3000e, band 2000e to 10000e, MPE = 1.5e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 15000 + 2.5$, corrected by $E0 = +0$.
weighing	12500.0	-0.5	7.5	7%	0.0 → PASS	PASS	R76-1 A.4.4; R76-1 3.5.1, Table 6	Error -0.5 g vs MPE ±7.5 g (load = 2500e, band 2000e to 10000e, MPE = 1.5e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 12500 + 2.5$, corrected by $E0 = +0$.
weighing	10005.0	1.0	7.5	13%	0.0 → PASS	PASS	R76-1 A.4.4; R76-1 3.5.1, Table 6	Error +1 g vs MPE ±7.5 g (load = 2001e, band 2000e to 10000e, MPE = 1.5e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 10005 + 2.5$, corrected by $E0 = +0$.
weighing	10000.0	0.0	5.0	0%	0.0 → PASS	PASS	R76-1 A.4.4; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±5 g (load = 2000e, band 500e to 2000e, MPE = 1e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 10000 + 2.5$, corrected by $E0 = +0$.
weighing	7500.0	0.5	5.0	10%	0.0 → PASS	PASS	R76-1 A.4.4; R76-1 3.5.1, Table 6	Error +0.5 g vs MPE ±5 g (load = 1500e, band 500e to 2000e, MPE = 1e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 7500 + 2.5$, corrected by $E0 = +0$.
weighing	6200.0	-0.5	5.0	10%	0.0 → PASS	PASS	R76-1 A.4.4; R76-1 3.5.1, Table 6	Error -0.5 g vs MPE ±5 g (load = 1240e, band 500e to 2000e, MPE = 1e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 6200 + 2.5$, corrected by $E0 = +0$.
weighing	5000.0	1.0	5.0	20%	0.0 → PASS	PASS	R76-1 A.4.4; R76-1 3.5.1, Table 6	Error +1 g vs MPE ±5 g (load = 1000e, band 500e to 2000e, MPE = 1e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 5000 + 2.5$, corrected by $E0 = +0$.
weighing	2505.0	0.0	5.0	0%	0.0 → PASS	PASS	R76-1 A.4.4; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±5 g (load = 501e, band 500e to 2000e, MPE = 1e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 2505 + 2.5$, corrected by $E0 = +0$.
weighing	2500.0	0.5	2.5	20%	0.0 → PASS	PASS	R76-1 A.4.4; R76-1 3.5.1, Table 6	Error +0.5 g vs MPE ±2.5 g (load = 500e, band 0e to 500e, MPE = 0.5e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 2500 + 2.5$, corrected by $E0 = +0$.
weighing	100.0	-0.5	2.5	20%	0.0 → PASS	PASS	R76-1 A.4.4; R76-1 3.5.1, Table 6	Error -0.5 g vs MPE ±2.5 g (load = 20e, band 0e to 500e, MPE = 0.5e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 100 + 2.5$, corrected by $E0 = +0$.
eccentricity	5000.0	0.0	5.0	0%	0.0 → PASS	PASS	R76-1 3.6.2.1, 3.6.2.2 / A.4.7; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±5 g (load = 1000e, band 500e to 2000e, MPE = 1e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 5000 + 2.5$, corrected by $E0 = +0$.
eccentricity	5000.0	0.5	5.0	10%	0.0 → PASS	PASS	R76-1 3.6.2.1, 3.6.2.2 / A.4.7; R76-1 3.5.1, Table 6	Error +0.5 g vs MPE ±5 g (load = 1000e, band 500e to 2000e, MPE = 1e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 5000 + 2.5$, corrected by $E0 = +0$.
eccentricity	5000.0	-0.5	5.0	10%	0.0 → PASS	PASS	R76-1 3.6.2.1, 3.6.2.2 / A.4.7; R76-1 3.5.1, Table 6	Error -0.5 g vs MPE ±5 g (load = 1000e, band 500e to 2000e, MPE = 1e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 5000 + 2.5$, corrected by $E0 = +0$.
eccentricity	5000.0	0.0	5.0	0%	0.0 → PASS	PASS	R76-1 3.6.2.1, 3.6.2.2 / A.4.7; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±5 g (load = 1000e, band 500e to 2000e, MPE = 1e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 5000 + 2.5$, corrected by $E0 = +0$.
eccentricity	5000.0	0.5	5.0	10%	0.0 → PASS	PASS	R76-1 3.6.2.1, 3.6.2.2 / A.4.7; R76-1 3.5.1, Table 6	Error +0.5 g vs MPE ±5 g (load = 1000e, band 500e to 2000e, MPE = 1e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 5000 + 2.5$, corrected by $E0 = +0$.
repeatability	7500.0	1.0	5.0	20%	-	PASS	R76-1 3.6.1 / A.4.10; R76-1 3.5.1, Table 6	10 weighings of 7500 g; spread of errors = 1 g vs allowed MPE = 5 g.
repeatability	15000.0	1.0	7.5	13%	-	PASS	R76-1 3.6.1 / A.4.10; R76-1 3.5.1, Table 6	10 weighings of 15000 g; spread of errors = 1 g vs allowed MPE = 7.5 g.
discrimination	100.0	5.0	5.0	-%	-	PASS	R76-1 3.8.2.2 / A.4.8	Added 7 g (1.4d) at 100 g; indication changed by 5 g (must change by at least d = 5 g).

Test	Load (g)	Error (g)	MPE (±g)	MPE Used	Naive rounding	Result	Clause	Explanation
discrimination	7500.0	5.0	5.0	-%	-	PASS	R76-1 3.8.2.2 / A.4.8	Added 7 g (1.4d) at 7500 g; indication changed by 5 g (must change by at least d = 5 g).
discrimination	15000.0	5.0	5.0	-%	-	PASS	R76-1 3.8.2.2 / A.4.8	Added 7 g (1.4d) at 15000 g; indication changed by 5 g (must change by at least d = 5 g).
zero_setting	0.0	0.0	1.25	0%	-	PASS	R76-1 4.5.2 / A.4.2.3	Error after setting = +0 g vs limit ±1.25 g (0.25e).
tare_setting	0.0	0.5	1.25	40%	-	PASS	R76-1 4.6.3 / A.4.6.2	Error after setting = +0.5 g vs limit ±1.25 g (0.25e).
temperature	100.0	0.0	2.5	0%	0.0 → PASS	PASS	R76-1 A. 5.3.1; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±2.5 g (load = 20e, band 0e to 500e, MPE = 0.5e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 100 + 2.5$, corrected by $E0 = +0$.
temperature	7500.0	0.0	5.0	0%	0.0 → PASS	PASS	R76-1 A. 5.3.1; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±5 g (load = 1500e, band 500e to 2000e, MPE = 1e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 7500 + 2.5$, corrected by $E0 = +0$.
temperature	15000.0	0.0	7.5	0%	0.0 → PASS	PASS	R76-1 A. 5.3.1; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±7.5 g (load = 3000e, band 2000e to 10000e, MPE = 1.5e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 15000 + 2.5$, corrected by $E0 = +0$.
temperature	100.0	0.0	2.5	0%	0.0 → PASS	PASS	R76-1 A. 5.3.1; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±2.5 g (load = 20e, band 0e to 500e, MPE = 0.5e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 100 + 2.5$, corrected by $E0 = +0$.
temperature	7500.0	0.0	5.0	0%	0.0 → PASS	PASS	R76-1 A. 5.3.1; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±5 g (load = 1500e, band 500e to 2000e, MPE = 1e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 7500 + 2.5$, corrected by $E0 = +0$.
temperature	15000.0	0.0	7.5	0%	0.0 → PASS	PASS	R76-1 A. 5.3.1; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±7.5 g (load = 3000e, band 2000e to 10000e, MPE = 1.5e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 15000 + 2.5$, corrected by $E0 = +0$.
temperature	100.0	0.0	2.5	0%	0.0 → PASS	PASS	R76-1 A. 5.3.1; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±2.5 g (load = 20e, band 0e to 500e, MPE = 0.5e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 100 + 2.5$, corrected by $E0 = +0$.
temperature	7500.0	0.0	5.0	0%	0.0 → PASS	PASS	R76-1 A. 5.3.1; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±5 g (load = 1500e, band 500e to 2000e, MPE = 1e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 7500 + 2.5$, corrected by $E0 = +0$.
temperature	15000.0	0.0	7.5	0%	0.0 → PASS	PASS	R76-1 A. 5.3.1; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±7.5 g (load = 3000e, band 2000e to 10000e, MPE = 1.5e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 15000 + 2.5$, corrected by $E0 = +0$.
temperature	100.0	0.0	2.5	0%	0.0 → PASS	PASS	R76-1 A. 5.3.1; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±2.5 g (load = 20e, band 0e to 500e, MPE = 0.5e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 100 + 2.5$, corrected by $E0 = +0$.
temperature	7500.0	0.0	5.0	0%	0.0 → PASS	PASS	R76-1 A. 5.3.1; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±5 g (load = 1500e, band 500e to 2000e, MPE = 1e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 7500 + 2.5$, corrected by $E0 = +0$.
temperature	15000.0	0.0	7.5	0%	0.0 → PASS	PASS	R76-1 A. 5.3.1; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±7.5 g (load = 3000e, band 2000e to 10000e, MPE = 1.5e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 15000 + 2.5$, corrected by $E0 = +0$.
temperature	100.0	0.0	2.5	0%	0.0 → PASS	PASS	R76-1 A. 5.3.1; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±2.5 g (load = 20e, band 0e to 500e, MPE = 0.5e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 100 + 2.5$, corrected by $E0 = +0$.
temperature	7500.0	0.0	5.0	0%	0.0 → PASS	PASS	R76-1 A. 5.3.1; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±5 g (load = 1500e, band 500e to 2000e, MPE = 1e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 7500 + 2.5$, corrected by $E0 = +0$.
temperature	15000.0	0.0	7.5	0%	0.0 → PASS	PASS	R76-1 A. 5.3.1; R76-1 3.5.1, Table 6	Error +0 g vs MPE ±7.5 g (load = 3000e, band 2000e to 10000e, MPE = 1.5e). Changeover method (A.4.4.3): $E = I + e/2 - \Delta L - L = 15000 + 2.5$, corrected by $E0 = +0$.
temperature_zero_drift	-	0.5	20.0	-%	-	PASS	R76-1 3.9.2.3	Zero indication drifted 0.5 g over 20 °C; allowed 1e per 5 °C = 20 g.

Test	Load (g)	Error (g)	MPE (±g)	MPE Used	Naive rounding	Result	Clause	Explanation
temperature_zero_drift	-	1.0	50.0	-%	-	PASS	R76-1 3.9.2.3	Zero indication drifted 1 g over 50 °C; allowed 1e per 5 °C = 50 g.
temperature_zero_drift	-	0.5	15.0	-%	-	PASS	R76-1 3.9.2.3	Zero indication drifted 0.5 g over 15 °C; allowed 1e per 5 °C = 15 g.
temperature_zero_drift	-	0.0	15.0	-%	-	PASS	R76-1 3.9.2.3	Zero indication drifted 0 g over 15 °C; allowed 1e per 5 °C = 15 g.
disturbance	-	0.0	5.0	-%	-	PASS	R76-1 T.5.5.6	Voltage variation: indication changed by 0 g under disturbance; a significant fault is anything above 5 g (1e). Recorded from the laboratory test sheet.
disturbance	-	5.0	5.0	-%	-	PASS	R76-1 T.5.5.6	Short-time power reduction: indication changed by 5 g under disturbance; a significant fault is anything above 5 g (1e). Recorded from the laboratory test sheet.
disturbance	-	0.0	5.0	-%	-	PASS	R76-1 T.5.5.6	Electrostatic discharge: indication changed by 0 g under disturbance; a significant fault is anything above 5 g (1e). Recorded from the laboratory test sheet.

Overall Verdict: FAIL



Verify online

Anita Sharma
Senior Reviewing Officer
Approved On: 2026-09-26T08:42:59+00:00

Scan the QR code or visit the link below to verify this report is genuine and unmodified. <http://localhost:8000/verify/bf5922834fd2de743764758c6bb81330b86c22461418e2ee39a35b61b988486c>
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